



May 18, 2026
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Proposed Challenges for
Erdos Alumni Club

*Challenge 1: **Supervised** Learning of Six-year Coastal Data-set*

We are seeking a collaboration to analyze the 6 year, continuous time series ($\Delta t = 5\text{min}$) of dissolved oxygen (DO) measurements obtained with our sensor technology which includes three water quality / hydrological variables and five meteorological parameters. Starting from an initial hypothesis, we have manually classified the DO events into distinct groups. The goal of this proposed project is to formalize the classification methodology.

*Challenge 2: **Modeling Sea-level Rise via Unsupervised** Learning of a Nine Parameter Temporally Resolved Multi-Year Data-set*

We are seeking a collaboration to develop predictive machine learning tools towards modeling the effects of climate change on coastal ecosystems subject to sea-level rise. To this end we seek to untangle the hydro-meteorological drivers of observed dissolved oxygen (DO) events via unsupervised learning of a multi-site, temporally resolved (at 5min resolution) data-set obtained in parallel with measurements of both weather and hydrological variables. Given the role of antecedent conditions which vary over diel, tidal, seasonal and yearly time-scales, the challenge is how to classify the observed DO events.

May 5, 2026
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Proposed Challenge for
Erdos Alumni Club

We are seeking a collaboration to analyze the 6 year, continuous time series ($\Delta t = 5\text{min}$) of dissolved oxygen (DO) measurements obtained with our sensor technology. To understand the drivers of DO in this coastal ecosystem, along with subsurface oxygen measurements our temporally resolved data set includes three water quality / hydrological variables and five meteorological parameters. Starting from an initial hypothesis, we have classified the DO events into distinct groups. The goal of this proposed project is to formalize the classification methodology.

This formalization of the DO event classification is the first step towards development of predictive machine learning tools to model the effects of climate change on coastal ecosystems subject to sea-level rise. Through Opti O₂'s sensor technology we envision having sufficient temporal resolution at all the relevant time scales, which range from minutes to years, to test the validity of proposed machine learning algorithms with site specific. This is especially relevant to proper classification of the DO events given observed role of antecedent conditions hydro-meteorological conditions.

We welcome the opportunity to work with the Erdos Alumni club, and hope this project will serve as the basis for integrating data science tools and rigorous analytic analysis with Opti O₂'s measurement technology..

I have attached on the next page the abstract of the presentation I gave recently at the Ocean Sciences Meetings.

Best,

Ruby

2026 Ocean Sciences Meeting, February 22 – 27, 2026, Glasgow, United Kingdom
 Oral paper: CB41A-03

**Evolution of a Former Freshwater Ecosystem to a Coastal Wetland
 Revealed by Six Continuous Years of High Frequency
 Groundwater Dissolved Oxygen and Hydro-meteorological Observations**

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The dynamics of lateral oxygen transport at the land-sea interface is critical to understanding the biogeochemical response of coastal ecosystems, especially in response to tidally driven surface water intrusion into the groundwater table. Using state-of-the art probes (Opti O₂) we continuously measured subsurface dissolved oxygen (DO) from 2019-2025 at a previously freshwater floodplain on the US Pacific coast exposed to seawater flooding following culvert removal. Our 5min resolution, allowed direct observation of two distinct classes of sediment oxygenation events, each with unique temporal signatures: 1) a series of *hot moments* or incursions of high DO tidal water, causing an abrupt rise in DO. These DO consumption events have an asymmetric temporal pattern with an inflection point and are coincident with step changes in subsurface salinity, water levels and temperature. 2) Individual *oxic pulses* which are temporally symmetric DO events, rising and falling at similar rates. These pulses typically occur either after significant precipitation or immediately before a series of tidally driven hot moments.

Co-located temporally resolved measurements of salinity, surface and groundwater levels, as well as meteorological parameters let us parse the drivers of these events, which may substantially alter coastal biogeochemistry. Wavelet analyses elucidates the interactions between DO, salinity, water level and temperature during oxygenation events followed by subsequent returns to anoxia. Information theory revealed DO's relationship with meteorological data and upstream conditions. We hypothesize that the evolving salinization of flood plain sediment, driven primarily by lateral transport of surface water following barrier removal, has impacted soil properties (for example the dominance of biotic vs abiotic processes) as well as ground-surface water hydraulic exchange. These changes are evidenced by dramatic change in the number, frequency of occurrence and magnitude of oxic pulses vs hot moments over the six years span of our study. We seek to utilize the temporal resolution of our long-term DO and hydro-meteorological data set to parse scale specific (ex: at hourly, diel, tidal, seasonal and yearly cycles) interactions and thereby provide insight on ecosystem response to sea-level rise.